

Table - students

Students_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

1. SELECT DISTINCT department
FROM Students;

department
IT
HR
finance

2. Average age per department

SELECT department,
AVG (age) AS avg_age
FROM student
GROUP BY department

Department	avg_age
IT	20.5
HR	22
finance	23

3. Department with more than 1 student.

```
SELECT department,  
       count * AS student_count  
FROM student  
GROUP BY department  
HAVING count > 1;
```

department	student_count
IT	2
HR	2

4.

```
SELECT *  
FROM students  
WHERE age BETWEEN 21 & 23;
```

Student ID	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5.

```
SELECT *  
FROM students  
WHERE (department IN ('IT', 'HR')) AND age > 22;
```

Student ID	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Section 2

6. SELECT department,
SUM AS total-credit
FROM courses
GROUP BY department
HAVING SUM > 5;

department	total-credits
IT	11

7. Courses not equal to 4 credits.

```
SELECT *  
FROM courses  
WHERE credits <> 4;
```

Course-ID	Course-name	department	credit
101	SQL Basics	IT	3
104	Excel	finance	2
105	Statistic	HR	3

8. Top 3 course by credits.

```
SELECT course id,  
        course-name,  
        credits  
FROM course  
ORDER BY credits DESC  
LIMITS 3;
```

Course ID	Course Name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

Section 3

9. SELECT
 max AS max_grade,
 MIN AS min_grade,
 AVG AS avg_grade
 FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.6

10. SELECT course_id,
 COUNT(*) AS enrollment_count,
 GROUP BY course_id;

Course ID	enrollment-count
101	1
102	1
103	1
104	1
105	1

Section 4.

11. SQL

Section 4

11. SELECT department,
SUM AS total_salary,
SUM AS total_bonus
FROM Salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
finance	70000	6000

12. SELECT department,
AVG AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG > 55000;

department	avg_salary
IT	61000
Finance	7000

13. SELECT employee_id,
name,
salary,
bonus,
salary + bonus AS total_compensation
FROM Salaries
WHERE salary + bonus > 6000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

SECTION 5

14. SELECT departments
 SUM AS total-budget
 AVG AS avg-budget
 FROM projects
 GROUP BY departments
 HAVING AVG > 70000;

department	total-budget	avg-budget
IT	27000	13500
finance	8000	80000

15. SELECT *
 FROM projects
 WHERE budget BETWEEN 50000 AND 120000
 AND department <> 'marketing';

project_id	projectname	department	budget
1	AI app.	IT	120000
2	payroll system.	finance	80000
5	HR portal	HR	50000